[INCH-POUND]
A–A–59789/13
8 March 2008

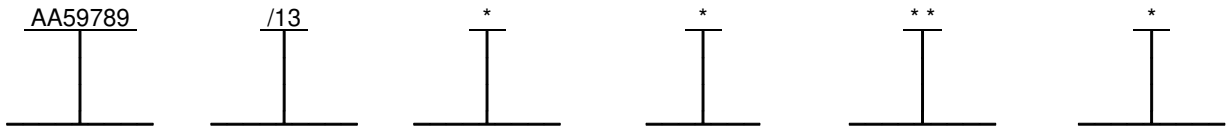
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 5 PIECE, FOR COLD PLATE APPLICATIONS, .375 X .375 INCH BODY SIZE, WITH TWO MOUNTING HOLES AND SCREW RETENTION, SCREW ACTUATED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring 5 piece card holders described herein shall consist of this document and the latest issue in effect of A–A–59789.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59789/13NP55R is the PIN for a nickel finished, 5.5 inch (139.7 mm) long card holder with a retained actuating screw. The card holder also features two counterbored holes for use with rivets and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in place providing high resistance to shock and vibration while providing maximum thermal transfer.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A–A–59789. Card holders shall also have a screw retention feature to prevent unintentional disassembly of the card holder.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59789.

Actuating screw hex drive socket. The across flats dimension for hex drive socket shall be .140 inch (3.60 mm) for mounting options “R” and “V” and .118 inch (3.0 mm) for mounting option “Y”.

Cold plate slot width. The recommended cold plate slot width to accommodate the circuit card assembly with attached card holder is .425 inch (10.80 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59789).

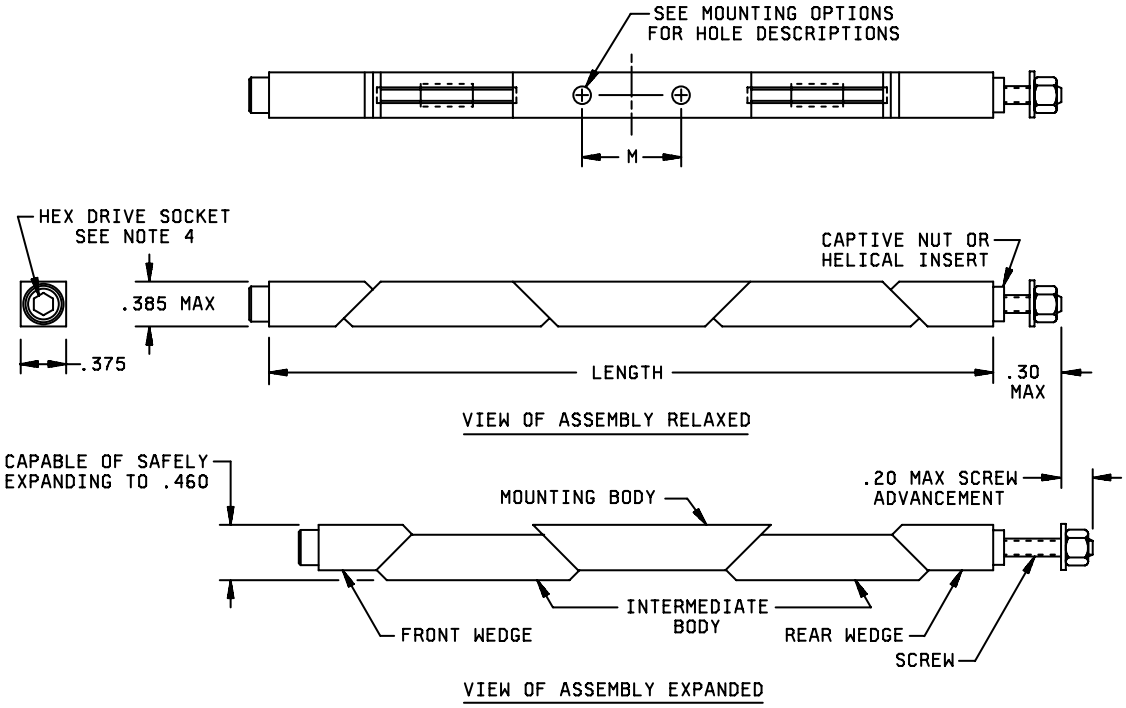
Installation torque. The recommended nominal installation torque is as follows: 20 inch-pounds (2.3 N-m) for assemblies of configuration “A” and “F” and 23 inch-pounds (2.6 N-m) for assemblies of configurations “J” and “N”.

Configuration. The configuration of card holders shall be as specified in table I. The details of a particular configuration consist of those on figures 1 and 2; and may include those on figures 3 or 4.

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TABLE I. Configurations.

Configuration	Applicable figures	Hardware options
A	1 and 2	No added hardware
F	1, 2, and 3	Lockwasher and flat washer
N	1, 2, and 4	Screw self-locking element
J	1, 2, 3 and 4	Lockwasher, flat washer, and screw self-locking element



Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.140	3.56	.30	7.6	.375	9.53	.380	9.65	.470	11.94

NOTES:
1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Tolerance for the mounting hole spacing is ±.005 inch (0.13 mm).
4. The across flats dimension for actuating screw hex drive socket shall be .140 inch (3.60 mm) or .118 (3.0 mm) for the corresponding mounting option (see hex drive socket details).
5. Configuration shown uses captive nut on rear wedge. Alternate design using threaded insert shall be acceptable.

FIGURE 1. Relaxed and expanded dimensions.

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Screw retention locknut (see figure 2). Card holders shall have a screw retention locknut intended to prevent unintentional disassembly of the card holder. The locknut shall be fabricated of a corrosion resistant material.

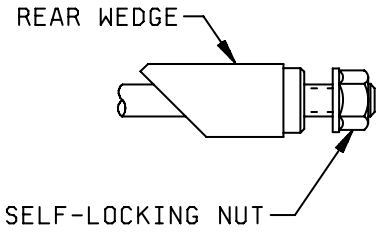


FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

Finish. The wedge body finish designator shall be as specified in A–A–59789. The finishes available for this CID specification sheet are as follows: “B” (black anodize), “P” (electrodeposited nickel), “H” (hard black anodize), or “D” (hard black anodize with dry film lubricant). When present, flat washers, lockwashers, and retention nuts shall be compatible with the specified finish.

Length, expanded, and relaxed dimensions. The length designator shall be as specified in A–A–59789. The lengths available for this CID specification sheet are listed in table II. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

TABLE II. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “length” ±.02 (0.5)	Dimension “M” ±.005 (0.13)
50	4.80 (121.9)	1.00 (25.4)
55	5.30 (134.6)	1.00 (25.4)
60	5.80 (147.3)	1.00 (25.4)
65	6.30 (160.0)	1.00 (25.4)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Mounting. Mounting designators “P” (indexing pins) “R” (rivet mount holes with counterbore and countersink), “S” (tapped 2-56 holes), “V” (tapped 4-40 holes), or “Y” (tapped metric M3 x 0.5 holes) shall be as specified in A–A–59789. See figure 1 for mounting hole spacing requirements. Card holders using mounting options “P” and “R” may be shipped unassembled.

Indexing pins. The two pins supplied for indexing pin mounting shall be Ø.062 x .040 inch (1.57 x 1.01 mm).

Rivet mount holes. The holes used for rivet mounting shall be .134/.139 inch (3.40/3.53 mm) diameter, countersunk 100 degrees by .190/.200 inch (4.83/5.08 mm) diameter through holes with a counterbore of .190/.200 inch (4.83/5.08 mm) diameter by .300 inch (7.62 mm) deep.

Rivets. This card holder uses rivet style B as specified in A–A–59789 when rivet mounting is used.

Configuration hardware options. Card holders can have the following hardware options: Lockwasher and flat washer, screw self-locking element, or a combination of both. See table I for the correct PIN configuration identifier.

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Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include configuration identifier “F” or “J” in the PIN (see table I).

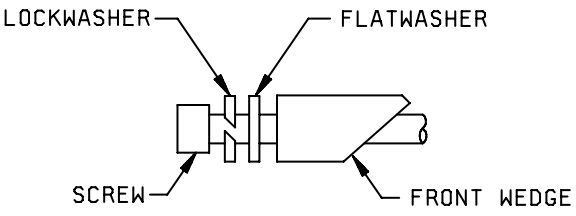


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock vibration. Card holders requiring a screw self-locking element shall include configuration identifier “J” or “N” in the PIN (see table I).

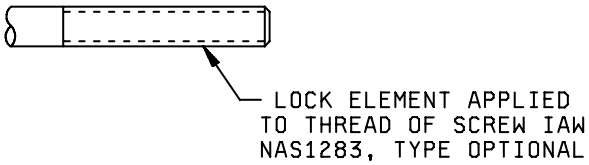


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include configuration identifier “J” in the PIN (see table I).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59789 – Holder, Electrical Card, Wedge Retainers, 5 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Other Publications

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA/NAS 1283 – Fasteners, Male Threaded, Self–locking.

(Application for copies should be addressed to the Aerospace Industries Association, 1250 Eye Street, NW, Suite 1200, Washington, DC 20005–3924 or at URL: http://www.aia-aerospace.org.)

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Ordering data. Ordering data is as specified in A–A–59789.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 E-mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet PINs supersede the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "50". See table IV for CID PIN construction using other available lengths for this CID specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. Commercial part number supersession data.

PIN designator AA59789/13	Vendor similar designator or type part number 1/ CAGE 52094	PIN designator AA59789/13	Vendor similar designator or type part number 1/ CAGE 52094
AB50V	A251–4.80T4R	AP50V	EN251–4.80T4R
AB50R	A251–4.80HR	AP50R	EN251–4.80HR
AB50S	A251–4.80T2R	AP50S	EN251–4.80T2R
AB50Y	MA251–4.80TM3R	AP50Y	MEN251–4.80TM3R
FB50V	WA251–4.80T4R	FP50V	WEN251–4.80T4R
FB50R	WA251–4.80HR	FP50R	WEN251–4.80HR
FB50S	WA251–4.80T2R	FP50S	WEN251–4.80T2R
FB50Y	MWA251–4.80TM3R	FP50Y	MWEN251–4.80TM3R
NB50V	A251–4.80T4LR	NP50V	EN251–4.80T4LR
NB50R	A251–4.80HLR	NP50R	EN251–4.80HLR
NB50S	A251–4.80T2LR	NP50S	EN251–4.80T2LR
NB50Y	MA251–4.80TM3LR	NP50Y	MEN251–4.80TM3LR
JB50V	WA251–4.80T4LR	JP50V	WEN251–4.80T4LR
JB50R	WA251–4.80HLR	JP50R	WEN251–4.80HLR
JB50S	WA251–4.80T2LR	JP50S	WEN251–4.80T2LR
JB50Y	MWA251–4.80TM3LR	JP50Y	MWEN251–4.80TM3LR

See footnote at end of table.

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TABLE III. Commercial part number supersession data. – Continued.

PIN designator AA59789/13	Vendor similar designator or type part number 1/ CAGE 52094	PIN designator AA59789/13	Vendor similar designator or type part number 1/ CAGE 52094
AH50V	HA251–4.80T4R	AD50V	HD251–4.80T4R
AH50R	HA251–4.80HR	AD50R	HD251–4.80HR
AH50S	HA251–4.80T2R	AD50S	HD251–4.80T2R
AH50Y	MHA251–4.80TM3R	AD50Y	MHD251–4.80TM3R
FH50V	WHA251–4.80T4R	FD50V	WHD251–4.80T4R
FH50R	WHA251–4.80HR	FD50R	WHD251–4.80HR
FH50S	WHA251–4.80T2R	FD50S	WHD251–4.80T2R
FH50Y	MWHA251–4.80TM3R	FD50Y	MWHD251–4.80TM3R
NH50V	HA251–4.80T4LR	ND50V	HD251–4.80T4LR
NH50R	HA251–4.80HLR	ND50R	HD251–4.80HLR
NH50S	HA251–4.80T2LR	ND50S	HD251–4.80T2LR
NH50Y	MHA251–4.80TM3LR	ND50Y	MHD251–4.80TM3LR
JH50V	WHA251–4.80T4LR	JD50V	WHD251–4.80T4LR
JH50R	WHA251–4.80HLR	JD50R	WHD251–4.80HLR
JH50S	WHA251–4.80T2LR	JD50S	WHD251–4.80T2LR
JH50Y	MWHA251–4.80TM3LR	JD50Y	MWHD251–4.80TM3LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.

TABLE IV. Example of PIN with available length designators.

PIN designator AA59789/13	Vendor similar designator or type part number 1/ 2/ CAGE 52094	PIN designator AA59789/13	Vendor similar designator or type part number 1/ 2/ CAGE 52094
NP50R	EN251–4.80HR	NP55R	EN251–5.30HR
NP60R	EN251–5.80HR	NP65R	EN251–6.30HR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.

2/ Other lengths are available on request.

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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2007–006

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